



GLOBALPAY

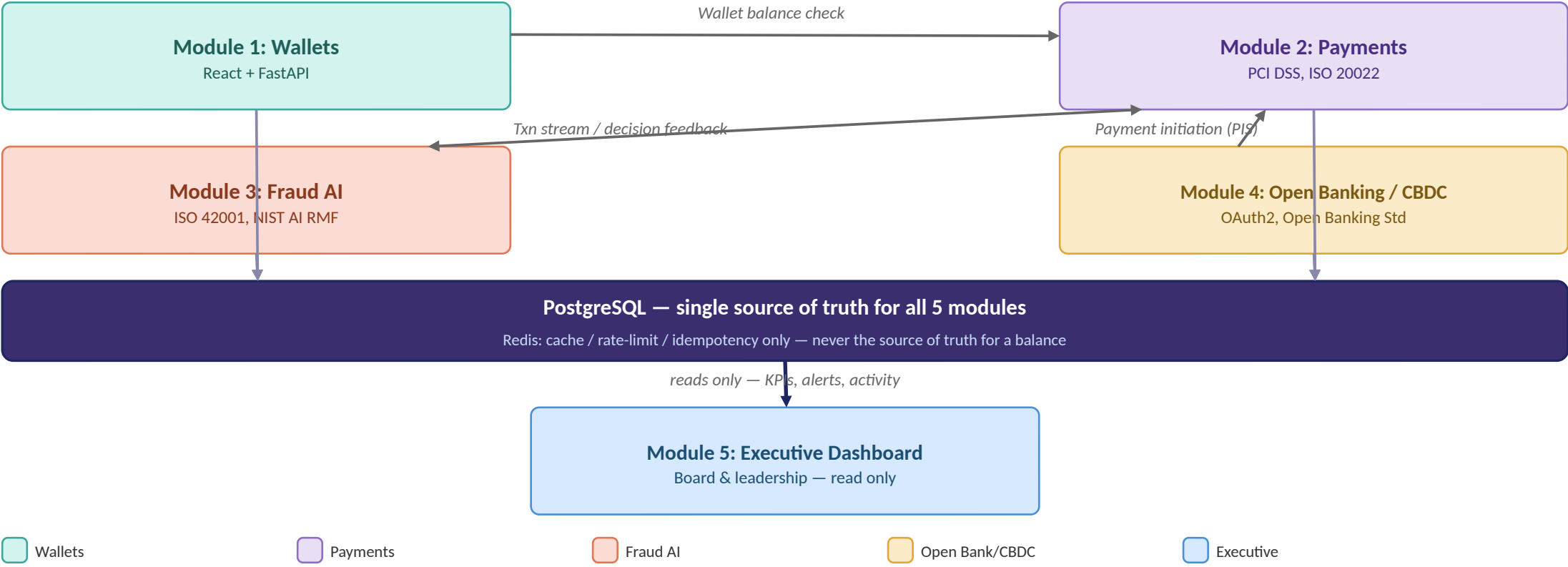
Section 8 — Complete Diagram Pack

All 17 mandatory diagrams, numbered exactly as listed in the exam brief

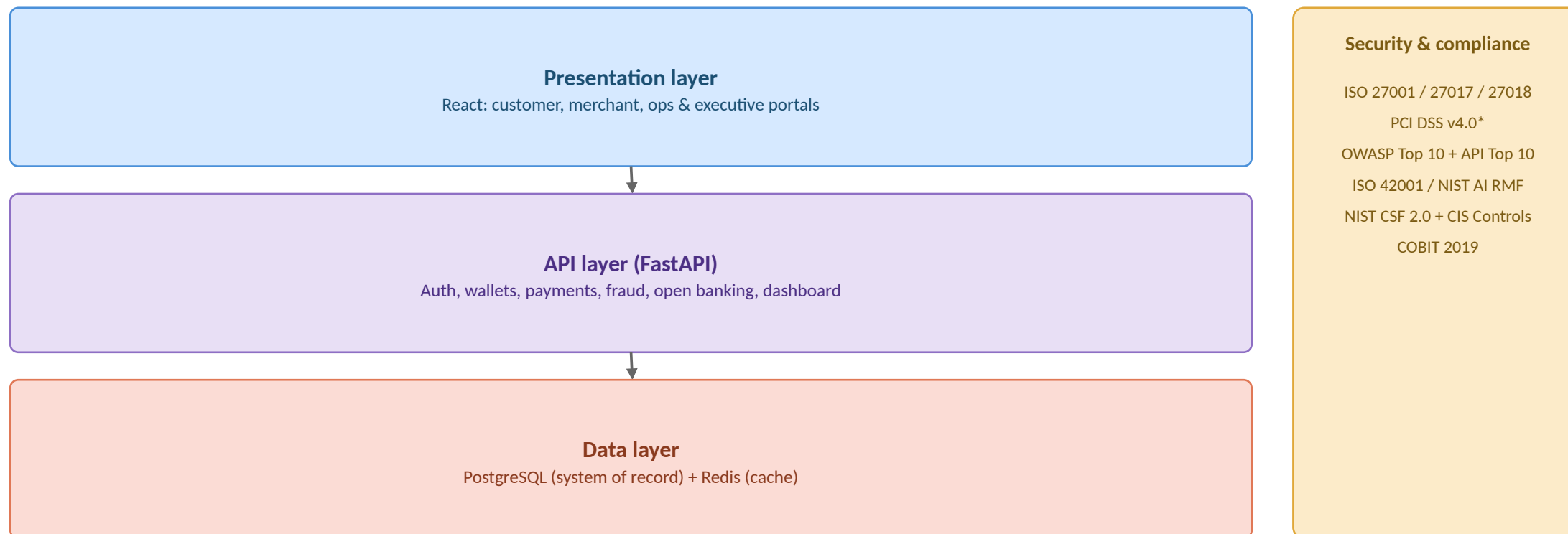
#	Diagram	Also in Stage 1 deck?
1	Enterprise Financial Technology Architecture	Yes
2	High-Level Solution Architecture	No - here only
3	Detailed System Architecture	No - here only (composite index)
4	Digital Wallet Architecture	Yes (Module 1)
5	Payment Processing Architecture	Yes (Module 2)
6	Open Banking API Architecture	Yes (Module 4)
7	AI Fraud Detection Architecture	Yes (Module 3)
8	CBDC Simulation Architecture	No - here only
9	Enterprise Network Architecture	Yes
10	Network Flow Diagram	Yes (combined in Module 11)
11	Data Flow Diagram (Level 0)	Yes (combined)
12	Data Flow Diagram (Level 1)	Yes (combined)
13	Digital Payment Workflow	Yes (combined in Module 2)
14	Fraud Detection Workflow	Yes (combined in Module 3)
15	Open Banking API Workflow	Yes (combined in Module 4)
16	Database ERD	Yes
17	Sequence Diagram	No - here only

G Enterprise Financial Technology Architecture

How the five modules connect



Modules 1-4 write to PostgreSQL; Module 5 reads only; only the named service calls above cross module boundaries



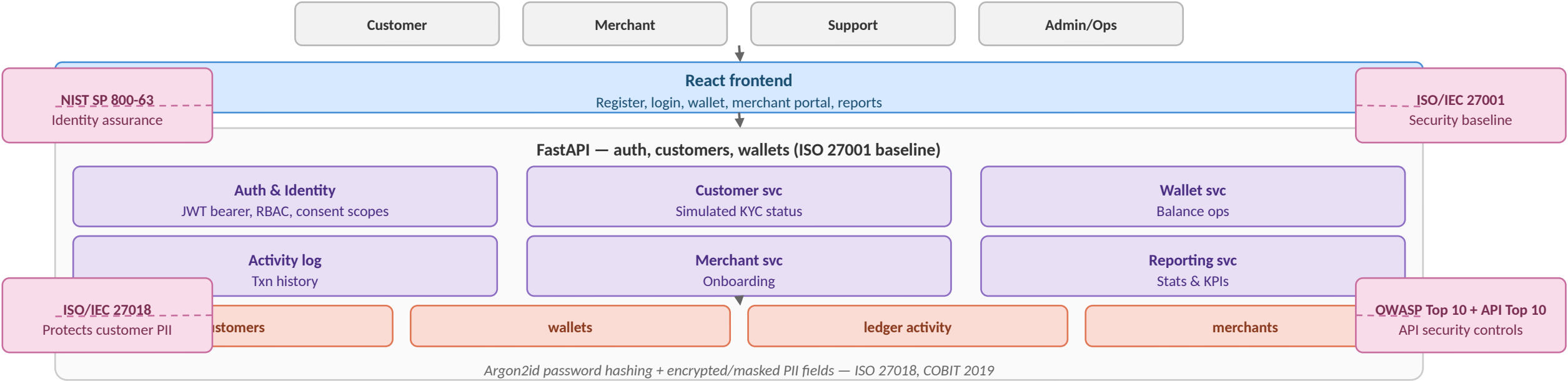
This is the 30,000-ft view — each module diagram (4-7) zooms into one API layer service

Composite reference — each module is fully detailed in diagrams 4-8

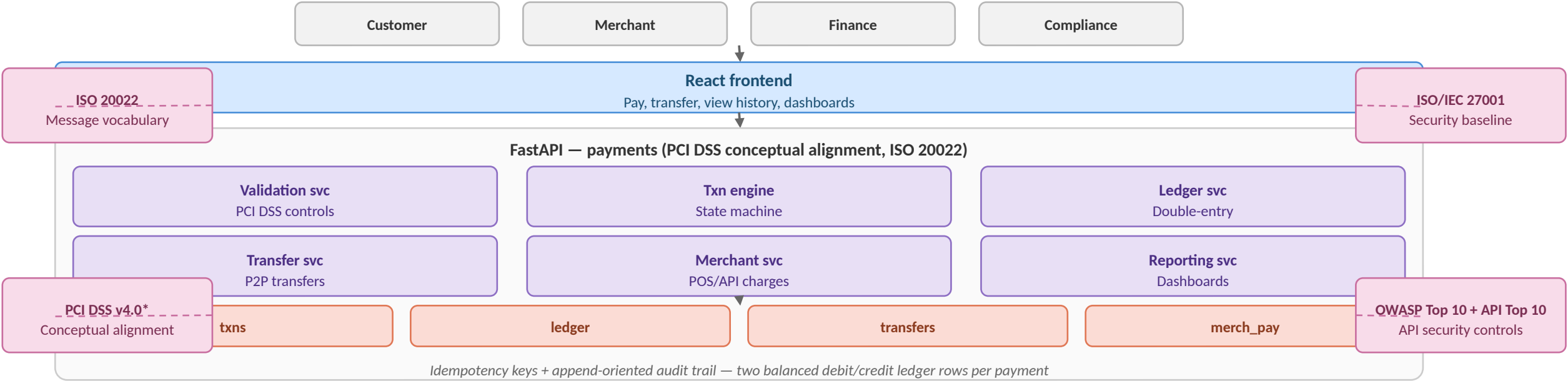
Module 1 Wallets	React UI -> FastAPI (auth, customers, wallets, merchants) -> PostgreSQL
Module 2 Payments	Validation, transaction engine, double-entry ledger -> ledger tables
Module 3 Fraud AI	Feature engineering -> Logistic Regression -> risk scoring -> human review
Module 4 Open Banking/CBDC	AIS/PIS APIs, CBDC API, activity monitoring -> separate CBDC ledger
Module 5 Executive Dashboard	Read-only aggregator across Modules 1-4 -> executive reports

This slide exists to index the detail, not duplicate it — see diagrams 4-8 for full architecture, framework badges, and data model per module

Registration, wallet creation, balance operations, merchant accounts

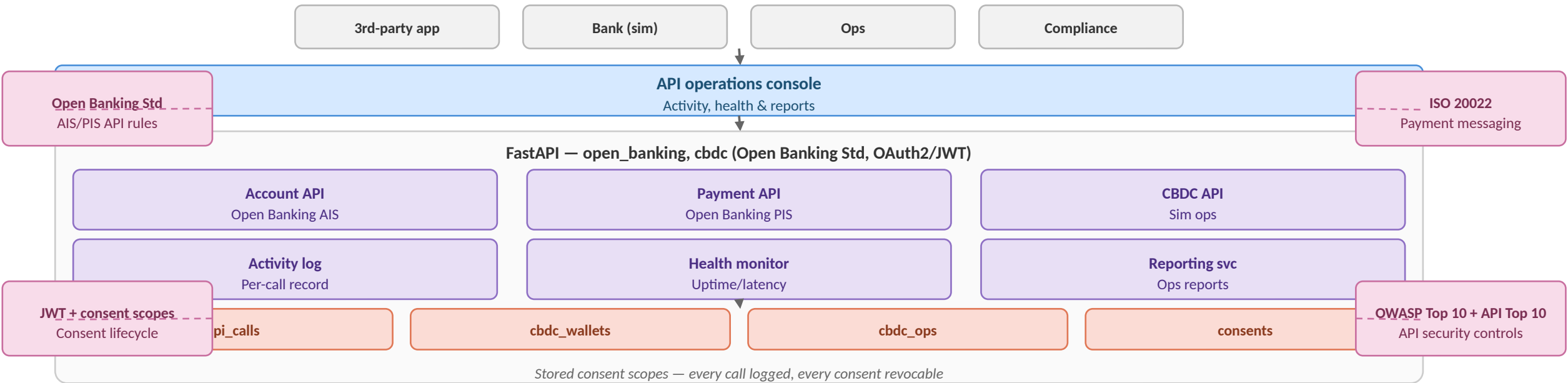


Validation, double-entry ledger, transfers, merchant charges



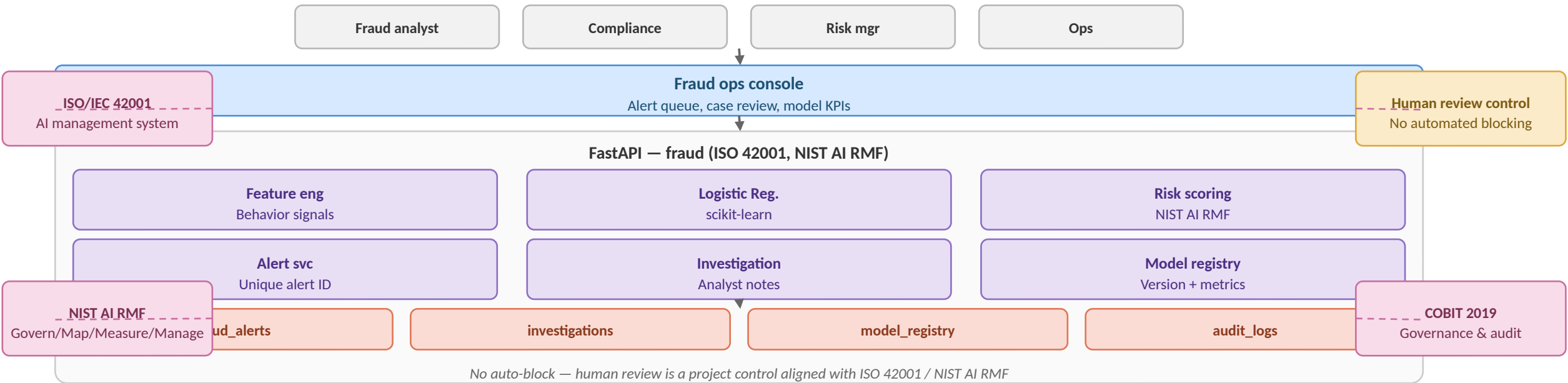
Direct link to Module 3 for fraud scoring — the only true module-to-module call besides Module 4's PIS

Account info & payment initiation APIs, CBDC wallet ops



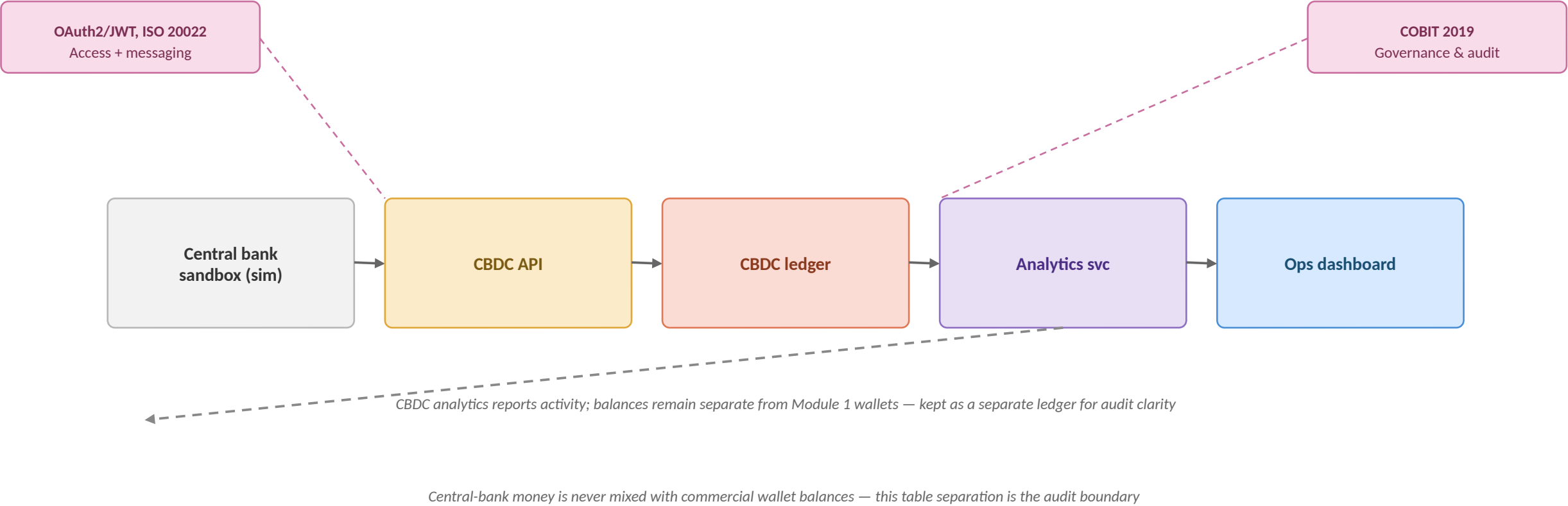
PIS calls Module 2's transfer logic directly rather than duplicating it — see diagram 8 for CBDC's separate ledger

Feature engineering, classification, mandatory human review

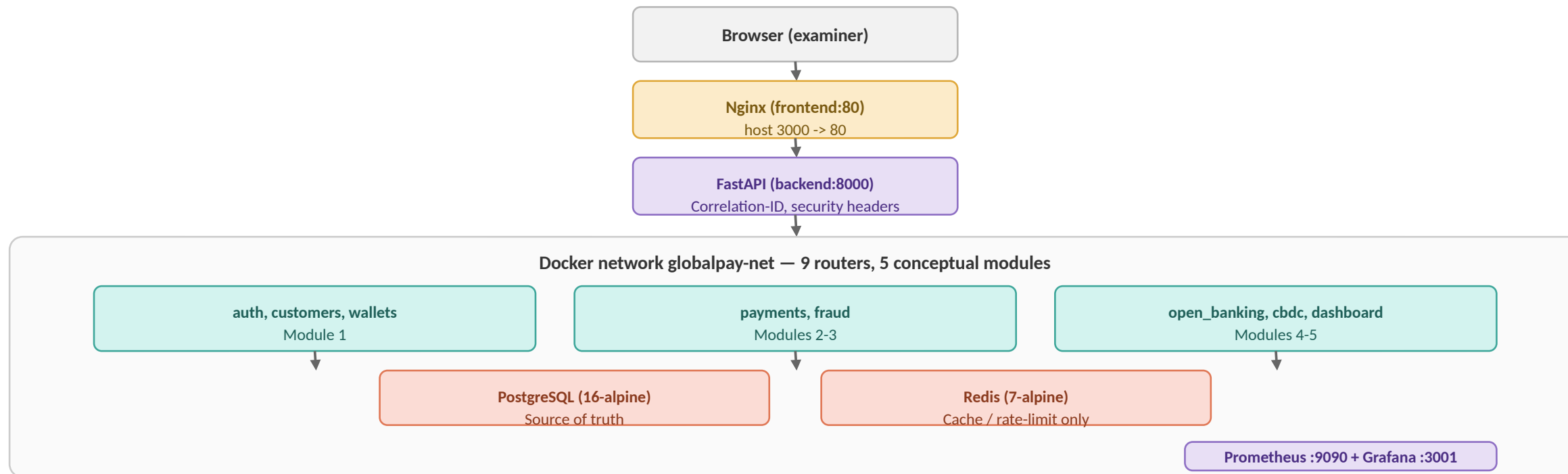


Surfaces to fraud/risk/compliance teams only — never shown to customers

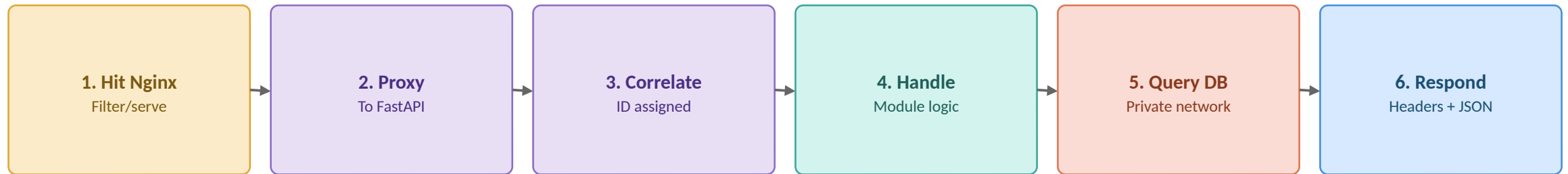
Kept separate from the commercial wallet ledger



compose.yaml — 6 containers on one Docker network (globalpay-net)

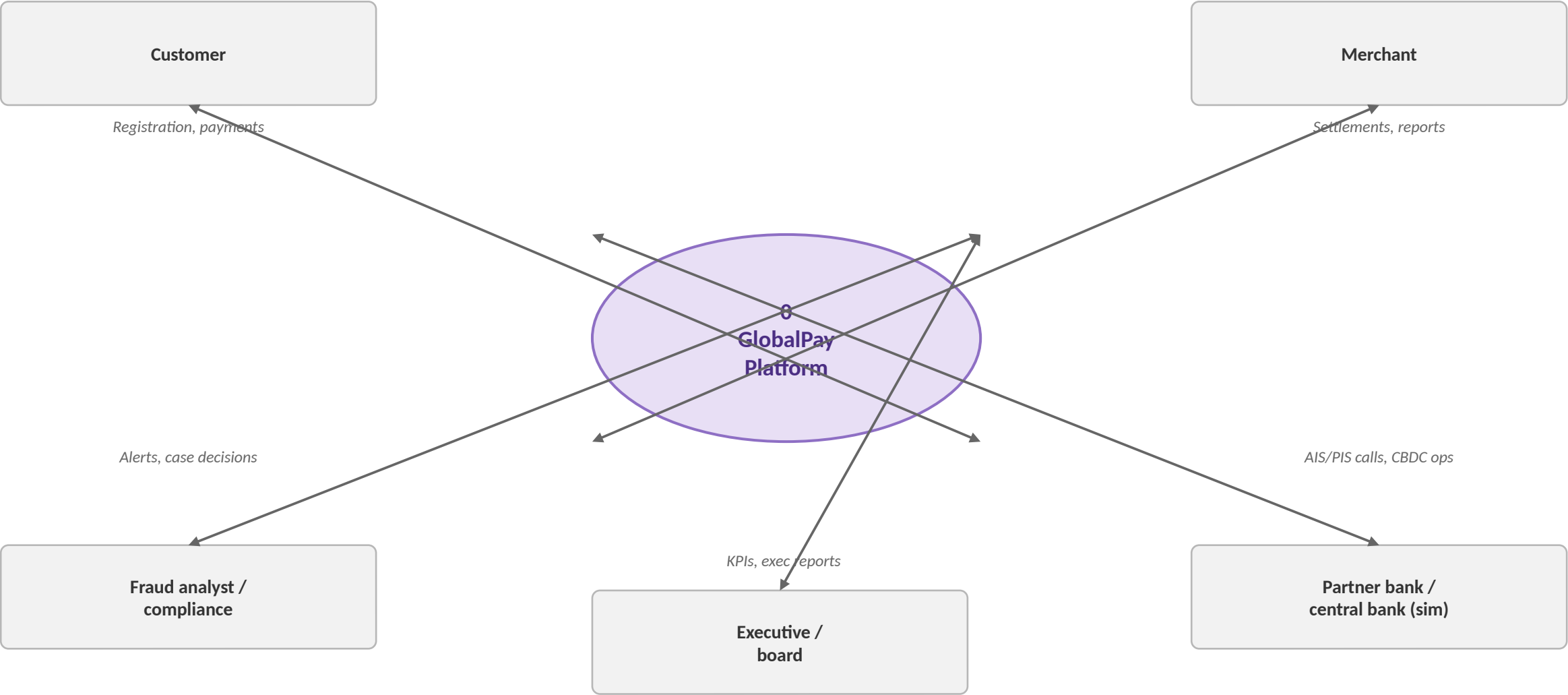


Redis never holds the authoritative wallet balance — Postgres is the single source of truth

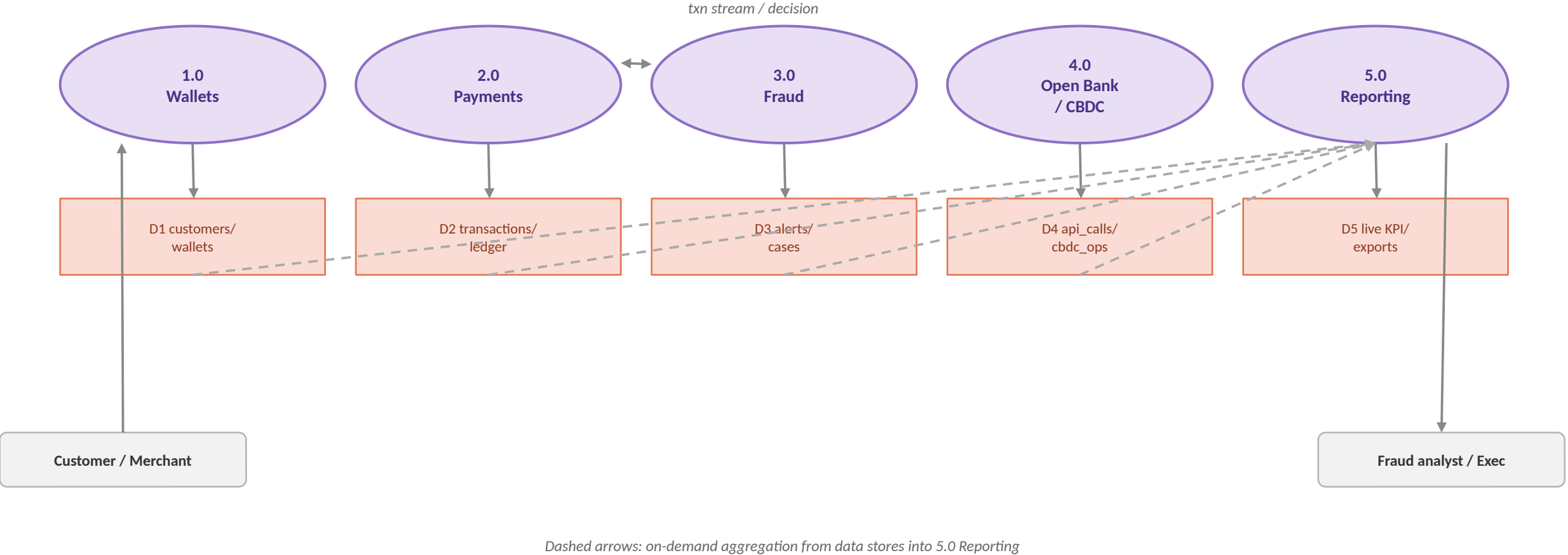


G Data Flow Diagram — Level 0 (Context)

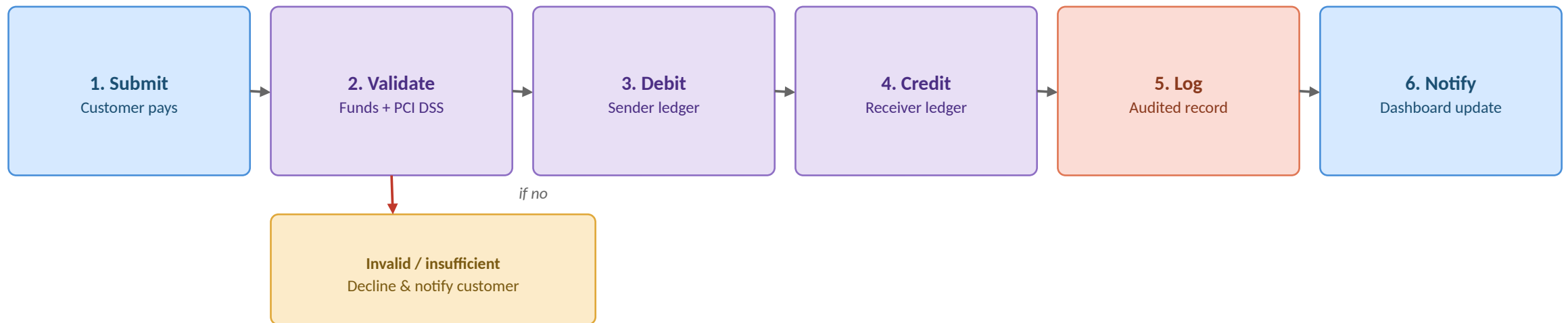
The platform as a single process, with its external entities

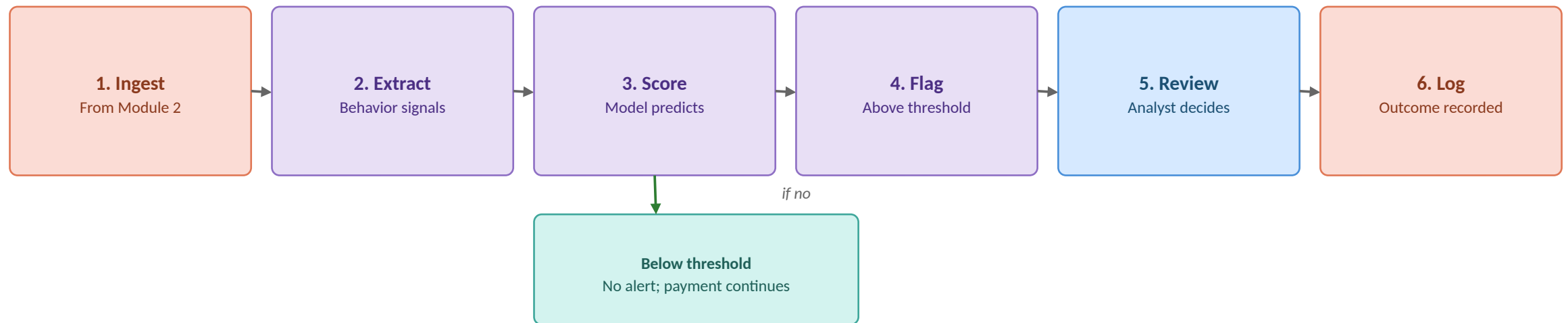


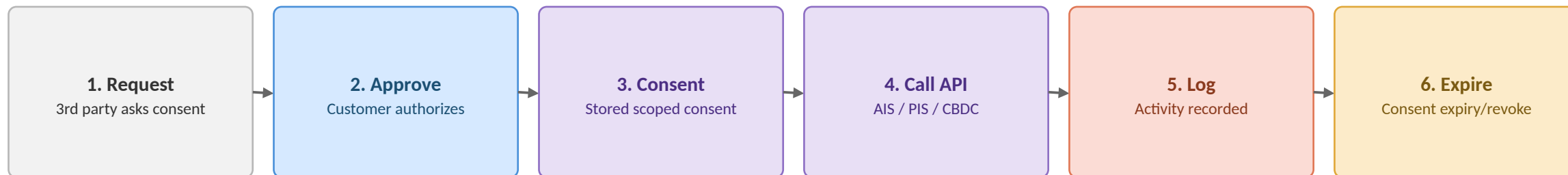
The platform decomposed into its five core processes



Validation branch shown explicitly



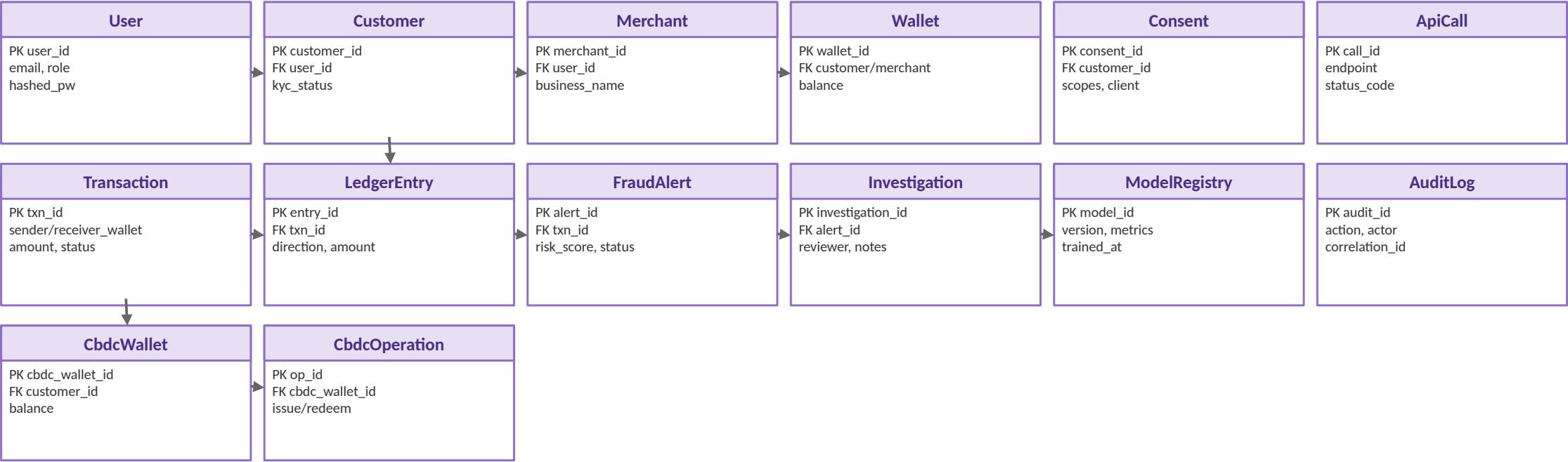




G

Database Entity Relationship Diagram (ERD)

14 tables from backend/app/models/entities.py



User -> {Customer, Merchant}; Customer -> {Wallet, Consent, CbdcWallet}; Merchant -> Wallet. Transaction -> {LedgerEntry, FraudAlert -> Investigation}. ModelRegistry and AuditLog are cross-cutting.

G Sequence Diagram — Registration Through Executive Reporting

17

End-to-end lifecycle across all five modules

